

Valorization and application of coffee husks extracts in cassava starch-based biodegradable packaging☆

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Highlights

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Coffee husk extract increased the antioxidant activity of cassava starch films.

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0.75 % extract improved tensile strength and increased film flexibility.

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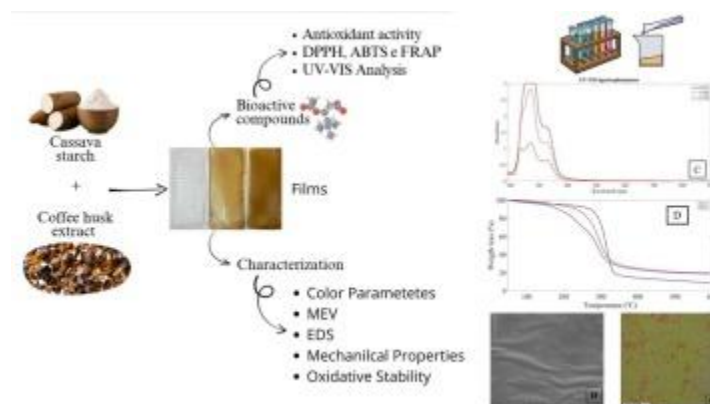
Addition of coffee by-product to biodegradable films improves mechanical, optical and functional performance.

Abstract

The growing demand for sustainable packaging with functional properties has encouraged the development of polymeric films incorporating bioactive compounds and the valorization of agro-industrial residues. In this study, active cassava starch-based films were produced (FC – without extract) and compared to films containing coffee husk extract at concentrations of 0.5 % (F5) and 0.75 % (F75). The aim of this work was to evaluate the effects of incorporating these compounds on the physical, thermal, mechanical, optical, and functional properties of the polymeric matrix. The Scanning Electron Microscopy (SEM) images revealed homogeneous surfaces with good extract dispersion, corroborated by elemental composition via Energy Dispersive X-ray Spectroscopy (EDS). Thermogravimetry (TGA) and Differential Scanning Calorimetry (DSC) analyses indicated higher stability, with degradation temperatures exceeding 300 °C. Regarding mechanical properties, tensile

strength increased from 10.31 MPa (FC) to 66.06 MPa (F75). Elongation at break also increased significantly in the extract-containing films, rising from 0.27 % (FC), characteristic of a rigid material, to 2.67 % (F5) and 3.28 % (F75), suggesting a plasticizing effect. The control film exhibited low absorption in the UV-Vis range, whereas the incorporation of the extract significantly increased absorption, improving UV-blocking capacity. Extract-enriched films demonstrated significant antioxidant activity, with values of F75 of 15.9 % (2,2-difenil-1-picril-hidrazil – DPPH), 44.3 % (2,2'-Azino-bis(3-ethylbenzothiazoline-6-sulfonic acid) - ABTS), 25.1 μmol Trolox/g (Ferric Reducing Antioxidant Power - FRAP), and 53.7 mg GAE/g (Total Phenolic Content - TPC). Additionally, oxidative stability was assessed using the Rancimat method, where the film containing 0.75 % extract (F75) significantly increased the induction period of soybean oil compared to the control, confirming the protective effect of coffee husk phenolics against lipid peroxidation. Therefore, the incorporation of coffee husk extract provided the films with multifunctional enhancements, reflected in improved structural, mechanical, optical, and functional properties, reinforcing their applicability as a viable and sustainable alternative for active packaging.

Graphical abstract



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Introduction

Packaging, in addition to protecting food from external conditions, also ensures the quality of the packaged products and the sanitary safety of food products, making transportation, storage, and distribution easier (Wyrwa & Barska, 2017). Due to the high demand, the massive use of petroleum-derived packaging leads to a negative environmental impact, as they are neither recyclable, compostable, nor biodegradable (Flórez et al., 2022).

As a result, there has been a growing interest in studies involving biopolymers, with chitosan, starch, and cellulose being the most used, and widely applied in the formulation of food packaging. Due to its availability, starch stands out as one of the most studied, in addition to being low-cost and highly biodegradable (Ayyubi et al., 2022).

The scientific community has focused attention on the application of active packaging, which has been widely studied as a strategy to enhance the functionality of biodegradable films. Active packaging is defined as packaging that includes active components that release or absorb substances to extend shelf life while maintaining the quality and sensory characteristics of the food. These can be classified into active removal systems (absorbers) and active release systems (emitters) (Flórez et al., 2022).

Recent studies have demonstrated that incorporating natural polymers, plant extracts, and nanoparticles into biopolymer matrices improves antioxidant and antimicrobial performance in food packaging films (Abdolsattari et al., 2020; Dadkhah et al., 2024; Erfani et al., 2023).

These combinations also contribute to enhanced mechanical strength, barrier properties, and controlled release of active substances, reinforcing their potential in sustainable packaging applications (Pirsa et al., 2023; Pirsa et al., 2024).

In this context, various natural compounds have been explored as active agents in biodegradable packaging formulations. A promising example is coffee, a beverage consumed worldwide, whose high consumption has generated an excessive amount of waste, such as the husk resulting from the thermal processing of coffee. In addition to being an abundant byproduct, coffee husk contains bioactive phytochemicals with antioxidant potential, such as polyphenols, phenolic acids, caffeine, tannins, and chlorogenic acids, among others (Hoseini et al., 2021). In this context, utilizing this waste promotes sustainable use, contributes to environmental protection and may facilitate the incorporation of bioactive compounds into the packaging (Liu et al., 2023).

Based on literature and the growing interest in natural bioactive additives, this study aims to develop and characterize a biodegradable active packaging incorporating coffee husk extract, a still underexplored agro-industrial byproduct in this application. The novelty lies in the use of hydroalcoholic extract from coffee husk in starch-based films, focusing on its influence on antioxidant activity and key physical and mechanical properties. The goal is that, in the future, they can be commercially applied, providing an upcycling alternative for waste and promoting a circular economy.

Materials and methods

The coffee husks from the dry processing of dry grains (a mixture of robusta and arabica coffee) and soybean oil, without antioxidants, were donated by COAMO Agroindustrial Cooperative (Campo Mourão, PR, Brazil). Ethanol (purity of 99.5 %, Dinâmica, Brazil) was used to obtain the extract, and cassava starch (Zaeli, Brazil) was used for the films. In the antioxidant analyses, the following reagents were used: Folin-Ciocalteu (purity ≥ 98 %, Sigma-Aldrich, USA), sodium carbonate (analytical

Characterization of the films

The Fig. 1 presents the films FC, F5, and F75.

Fig. 1 illustrates the color change in biodegradable cassava starch films with the incorporation of different concentrations of coffee husk extract. The control film (FC), on the left, shows a translucent color, typical of the absence of additives. With the addition of 0.5 % (F5) and 0.75 % (F75) of extract, located in the center and to the right, respectively, an increase in the intensity of the yellowish color is observed.

This chromatic change was

Conclusion

The incorporation of coffee husk extract into cassava starch-based biodegradable films proved to be an effective strategy for enhancing the functional properties of these materials. The enriched films showed a significant increase in antioxidant capacity, particularly the film containing 0.75 % extract, which reached values of 53.70 mg GAE/g of phenolic compounds, 44.31 % ABTS radical scavenging, and 25.15 μmol Trolox/100 g in the FRAP assay, confirming the antioxidant potential of the phenolic

CRedit authorship contribution statement

Gabriele Paola da Silva Francisco: Investigation, Formal analysis. **Valéria Maria Costa Teixeira:** Investigation, Formal analysis. **Grazielle Oliveira:** Investigation, Formal analysis. **Matheus Samponi Tucunduva Arantes:** Investigation, Formal analysis, Data curation. **Washington Luiz Esteves Magalhães:** Writing – original draft, Investigation, Formal analysis. **Vanessa Carvalho Rodrigues:** Investigation, Formal analysis. **Angela Maria Gozzo:** Supervision, Investigation. **Evandro Bona:** Investigation,

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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